

The Future of Work with Enterprise IS: Low-Code/No-Code Platforms and Democratization of Development

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1. Execution Summary

This paper discusses the the impact of LCNC platforms on businesses, information systems, and the future of work. The ability to create applications without extensive knowledge of coding via LCNC technologies and software is referred to as citizen development. LCNC platforms and technologies streamline the process of developing applications and fosters collaboration between the IT department and other business units which, as noted in the analysis, boosts digital transformation, agility, and cost efficiency, and cross department collaboration (Forrester, 2025). Increasing productivity, accelerating time to market, and productivity in various sectors are some of the quantifiable metrics noted in the case studies sociology literature. Yet, some of the aspects that are difficult to resolve include governance, integration in an enterprise system, and data security. The paper states that LCNC platforms and technologies are governance sustainable which foster a culture of innovation with trained robust frameworks and controls. Transformative governance, compliance, and culture are necessary for investment on workforce upskilling to facilitate the scalability of the organization. Finally, LCNC platforms have been noted in the paper to facilitate transformational changes in the enterprise software development process and digital innovation that is accessible.

2. Introduction

An enterprise information system (EIS) is pivotal in organizing functions of data engineering, operation, and innovation within an enterprise. Software development of the past century was done in a structured way, in a series of steps using the Waterfall model of development, and was sequential and rigid in nature. This was a robust approach to problem solving in a stabilized environment, and the approach, if not properly adapted, often resorted to waste. The year in the early 2000's emerged a paradigm shift to Agile methodologies that introduced abstraction development, embraced teamwork, and focused development to the end-users which set a precedence to flexibility and responsiveness.

Non-technical users within an organization can easily utilize LCNC (Low-Code/No-Code) platforms thanks to the paradigm shift they were built upon. Software and system applications within an organization can now be built within a short time using templates that are automated, and workflows that are built using graphic diagrams. As the need to be innovative during digital transformation, they gap the need for agility, efficiency, and innovation in enterprise systems.

This study explores the impact of LCNC platforms on Enterprise Systems by disregarding the barriers to development and development speed on digital transformation. It studies the platforms' constituent technical architecture, business impact, and governance challenges, and outlines strategies for sustainable adoption. The emphasis is on understanding what businesses can do to optimize productivity, user enablement and compliance with the help of LCNC technologies to energize a more agile and open future of work.

3. Background / Literature Review

The past decade in software development has been transformative as it moved from the conventional styles to Agile frameworks, and then Low-Code/No-Code (LCNC) platforms. Development in the past, using the waterfall model, had numerous requirements which were linear and sequenced, thus development cycles were ever too prolonged subsequently resulting in delays and extra costs (Hoisl & Abran, 2016). However, this model works under the ever-changing nature of the business and therefore innovation and responsiveness is stunted.

Agile approaches, which were developed in the early 2000s, focused on greater flexibility and speed from iterative development and collaboration (Malla, 2025). Not only did it change the work culture of teams, it had also brought

about greater changes on the culture bias of businesses to improve on customer service and digital transformation.

These foundations allowed the emergence of LCNC platforms as paradigm shift technologies enabling non-technical business users or citizen developers to create and use applications without any coding proficiency (Lu & Siau, 2021). This is the early phase of the democratization of innovation, during which business specialists will be able to use their expertise to create functional software solutions capturing the multifaceted domain of their enterprises. By removing the boundaries on software development, organizations can innovate faster, expand their scope of participation, and introduce automation to key newly created workflows, alongside robust governance to minimize risk and ensure quality, compliance, and control (Singh & Garg, 2025). Preceding the LCNC platforms, rapid application development tools centered on shortening build cycles and ease of use of scripting languages. The term "low-code" was first used in 2014 and is attributed to Forrester (Richardson, 2014), with OutSystems, Mendix, and Microsoft Power Apps as the front runners of the industry.

Once more, it has been demonstrated how LCNC focuses on funnels for cross-pollination to lift the business agility digital transformation efforts system wide while unlocking capacity on the bridge between business digital transformation initiatives. On the other hand, the governance challenges remain, like how to keep control for fostering citizen development, the shadow IT governance issue, and the enterprise-grade data security, scalability and integration issue (Loveth & Vivian, 2025). It has also been pointed out that, as LCNC scales in enterprise environments, the governance edges gets more emphasized (Rymer et al, 2017). Additionally, to keep the balance between LCNC and Corporate IT speed and control, it is likely that the governance frameworks in place will provide the required corporate IT standards that will enforce the LCNC cycle use and more so, Corporate IT resource use for sustained pass-along capture on the balance. It has been pointed out that the governance frameworks provide a path, as it is desired, to the success of the LCNC adoption in enterprise settings (Rymer et al, 2017). More so, oversights in application landscape fragmentation as well as compliance-policing gaps have been noted in the context of democratizing governance for compliances that need More So for control Singh and Garg 2025). Thus, more insightful introspection into elastic control frameworks and the platforms themselves, along with desired cultural governance, will be paramount for optimized LCNC distributed enterprise adoption.

The conclusions drawn in this synthesis on earlier works and emerging trends suggest that LCNC platforms continue to shape software development by enabling inclusivity and efficiency, while also needing further investigation and maneuvering to bring about new research and strategy on scalability, governance, and integration (Loveth & Vivian, 2025; Lu & Siau, 2021; Singh & Garg, 2025).

4. Analysis and Discussion

Like most technologies, digital transformation is experiencing a shift with the advent of Low-Code/No-Code (LCNC) platforms which eases the process of application development. According to Research and Markets (2023), the LCNC market is projected to grow to \$65 billion by 2027, which is a 32.2% CAGR. According to Gartner (2023), by 2025, 65% of all application development will be done through these platforms which will simplify and 'democratize' the coding process to a more user-friendly, visual format.

Architecture and functionality of technology

Through digital architecture frameworks, low-code/no-code (LCNC) platforms abstract intricate systems of programming capabilities into concepts of colossal simplicity displayed in high-definition graphics. Such abstraction enables users, even those who lack expertise in coding, to develop automation solutions for processes and create business applications. The core technology that underlies it partitions into four essential components, or layers:

1. Presentation layer: user-defined drag-and-drop interfaces with other extensive outliners.
2. Logic Layer: business rules and workflows are controlled through technique visualization and scripting.
3. Data Layer: information is securely stored, modeled and retrieved.
4. Integration layer: Outbound interfaces and pluggable APIs with systems like SAP, Salesforce, and Microsoft Office 365.

This architecture allows users to decide on the interface-data integration sequence of their solutions. From a technical standpoint, one of the core features is scalability, which is achieved through cloud-native frameworks and autoscaling. Through the deployment and expansion of the paperless factory within 15 departments as a business model,

Schaeffler was able to decrease their ramp-up time with the help of Mendix (2022). An important shift in LCNC platforms is the integration of core data governance systems, which streamline compliance for customers and vendors to security regulations such as SOC 2 and HIPAA.

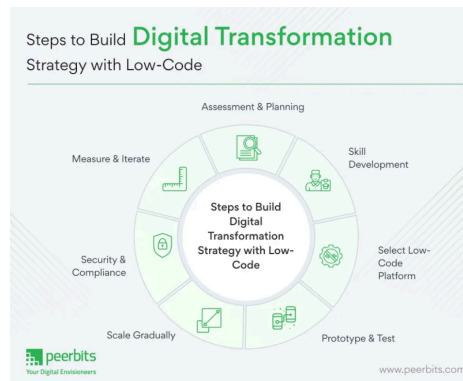


Fig. 1. Seven steps to construct a digital transformation strategy with optimized low-code tools solutions for assessment, skill assessment, tool selection, prototyping, scaling, security, iteration.

Enterprise case studies and applications

The productivity impact of LCNC is notable within various business domains.

- The Microsoft Power Platform in Practice: At Pacific Gas & Electric, more than 300 complex solutions have been created, and their AI chatbot saves \$75 million every year, while helpdesk costs shrank by \$1.1 million (Microsoft, 2025). In the past two years, Cineplex has decreased resolution times from 15 customer service query minutes to below thirty seconds, saving more than \$1 million and 30,000 hours (Microsoft, 2025).
- With Mendix, Schaeffler China developed a paperless factory solution that cuts down on 30% of development resources and integrates SAP, MES, and CMS systems without a hitch (Mendix, 2022). At Toyota Motor Corporation, it is estimated that the OutSystems Sheet Metal System used in over 70 projects reduces working hours by over 30% in comparison to traditional development systems (OutSystems, 2022).
- Nsure.com has automated its insurance comparison platform using the Microsoft Power Platform. Generative AIs have also been able to reduce the time spent on manual processing by 60% along with 50% of the costs during generated AIs processing. (Microsoft, 2025).

Strategic Advantages and Business Impact

1. Productivity and Cost Optimization

- Adopting lower-code and no-code (LCNC) platforms offers significant productivity and cost advantages. Businesses report development speeds up to ten times faster than traditional methods in order to respond to market changes (Forrester, 2022). Such extreme acceleration in development times is the main reason for the high ROI. Forrester (2022) shows that companies investing in modern integration tools and methods achieved a ROI of 176% within 3 years, 40% less downtime, and 33% to 67% less time for projects. The business is able to save by diminishing the need for skill of engineers, lowering the maintenance expenses, and faster launching of the product to the market. It is estimated that the use of LCNC tools increases the business value by approximately \$4.4 million within 3 years (Project Management Institute, 2023) from the absence of 2 IT developer positions.

2. Human Capital and Workforce Transformation

- Perhaps the most dramatic aspect of LCNC is the transformation of workforces. The LCNC platform is expected to be used by 80% of people employed as developers that work outside formal IT departments by 2026 (Gartner, 2023). IT professionals are the ones that pay the most price as LCNC allows business professionals to become creators at an unprecedented scale.
- More and more, technical professionals are facilitating, architecting, and guiding citizen development initiatives instead of doing every step by themselves. This transition presents new career pathways for individuals with deep domain expertise alongside application development skills.

Analysis of Challenges and Strategic Positioning

There are many possible advantages, and organizations, on the other hand, have to overcome many possible

disadvantages. A SWOT analysis indicates some of the important dynamics:

- Rapid development and industrialized scaling in the cloud, empowerment of the workforce, and tight integration of disparate systems.
- 37% of organizations as of 2023 (KPMG, 2023) are worried about vendor lock-in and the inability to customize for complex applications.
- New hybrid business models on the global scale, AI development, IoT, and market expansion.
- 25% of enterprises as of 2023 (KPMG, 2023) recognize the need to address data security vulnerabilities, while shadow IT is a real risk lacking proper controls.

The article concentrates on external constraints such as the GDPR and HIPAA, as well as market and workforce pressures, which together accelerate LCNC. Without the accompanying organizational change management plan for LCNC, success will be elusive.

Future Trends and Strategic Implications

The integration of LCNC with AI technology is an integral part of its future evolution. According to Gartner, by 2026, 75% of new enterprise applications will be developed on AI-led LCNC platforms. Through these platforms, users interface with the system using natural conversations to specify the requirements and the system, using generative AI, builds the software applications. Moreover, LCNC platforms are instrumental to Industry 4.0 where they help to integrate IT and OT. With the streamlined connection to the industrial systems, manufacturing and logistics firms are able to develop IoT dashboards and predictive maintenance applications in record time.

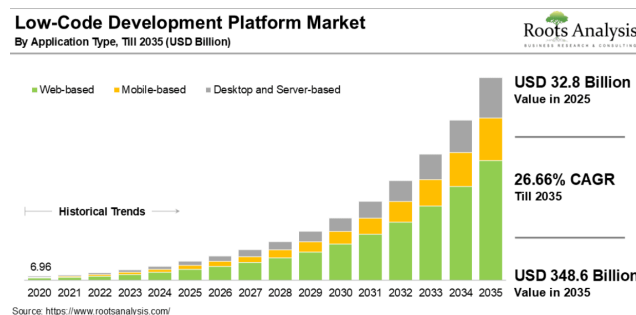


Fig.2 Low-Code/No-Code platforms operational in the market between 2020 and 2035 with the projected growth and adoption milestones.

5. Recommendations

To fully leverage the transformative capabilities of Low-Code/No-Code (LCNC) platforms, organisations should pursue an integrated approach that includes governance, workforce initiatives, and culture. For the first element, organisations will need to formalise governance frameworks that clearly delineate the roles and responsibilities of citizen developers while ensuring adherence to corporate IT and compliance frameworks. Governance frameworks should provide the necessary guardrails to prevent data fragmentation and unsanctioned disbursement controls (Forrester, 2025).

Second, organisations will need to ensure that there are systems of building capabilities and systems of scaffolding for training. While IT staff will provide training that scaffolds system development and helps to create IT guardrails, certification programmes will enable non-IT workers to create systems with digital solutions. Establishing Low-Code/No-Code Centres of Excellence (CoEs) offers organisations another opportunity to promote best practices and help achieve consistency and enterprise-scale support.

To cement installed LCNC systems, organisational culture will need to change. Cultures of responsible experimentation and innovation, along with collaboration and cohesion between business and IT functions, are crucial for embedding LCNC into the organisational digital mindset. When training, governance, and culture intertwine, organisations realise sustained empowered workforce, heightened agility, and rapid innovation cycles. All the benefits of democratised software development will be realised in the future of work.

6. Conclusion

LCNC platforms are changing enterprise software development by democratising innovation and closing the gap between business and IT. LCNC's improved agility and lower software development costs are accompanied by the ability of non-technical personnel to digitally transform portions of business processes. LCNC early adopters exhibit the integration of governance, training of the relevant workforce, and adaptability of the organisational culture, which Collins and his associates (Forrester, 2025) describe as the 'holistic approach' to scalable and compliant LCNC use.

Insights show that the value of LCNC implementations undermining organisational governance investments, such as shadow IT and issues of data privacy and interoperability, may persist indefinitely. Governance of citizen development depends on the establishment of organisational CoEs as described in the LCNC literature.

The future development of AI-enabled LCNC tools is expected to advance application design and development processes, as well as rapid deployment functions. Further work is required on the international LCNC oversight integration, governance, and the citizen development potential of generative AI. Aligning schema, processes, and tools, enterprises will use LCNC to enhance agility and inclusiveness within the innovation frameworks of their digitally transformed business processes.

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8. Appendix / Logbook

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Project Title: The Future of Work with Enterprise IS: Low-Code/No-Code Platforms and Democratization of Development

Date & Time	Task & Tools Used	Keywords/Prompts
9/10/2025 04:00pm	google	Example of Low-Code/No-Code Platforms Transforming Enterprise Development
11/10/2025	gemini 2.5 pro	Give any references related to Low-Code/No-Code Platforms Transforming Enterprise Development for learning purpose
11/10/2025	gemini 2.5 pro	Summarize task, what should us done in each section for academic writing
11/10/2025	gemini 2.5 pro	Separate task into two person
13/10/2025	gemini 2.5 pro	Help me check whether the execution summary and introduction is logically correct or not
16/10/2025	gemini 2.5 pro	Help me check whether the background, analysis and discussion is logically correct or not
18/10/2025	gemini 2.5 pro	Help me check whether recommendation and conclusion match with previous statement and logically correct or not
19/10/2025	gemini 2.5 pro	Above is my full documentation, help me last check whether any logically incorrect information.